



MASTERCARD ADVISORS • PERFORMANCE ANALYTICS

Autobench

The Precision Compliance Engine

Next-generation benchmarking. Minimal distortion. Absolute control.

Pedro Chagas
Associate Managing Consultant, ACS
Performance Analytics

Compliant benchmarking forces a trade-off between privacy and market truth.

Autobench resolves it mathematically, not manually.



THE RISK

Control 3 constraints

- Strict concentration caps (no peer above 25% share).
- Minimum participant thresholds per category.
- Zero-tolerance regulatory environment.
- Manual weighting silently destroys data fidelity.



THE GOAL

Market reality

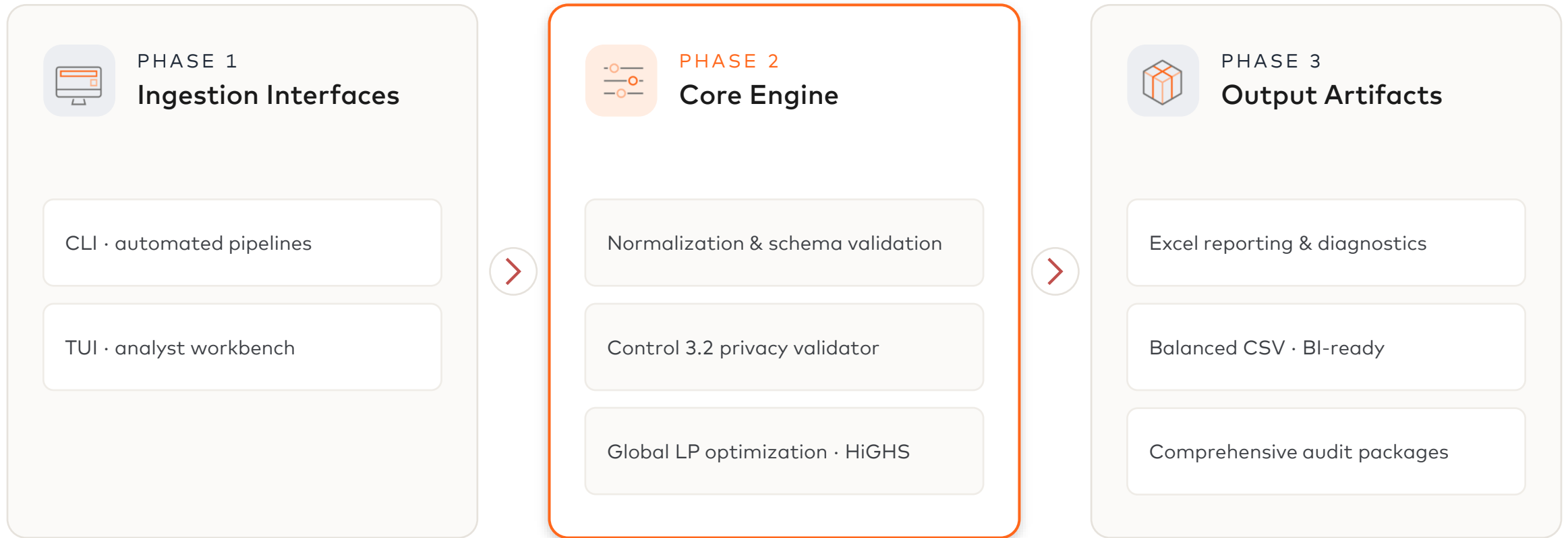
- Maintain exact market positioning.
- Preserve rank integrity across every dimension.
- Deliver actionable merchant and issuer performance.



The solution: Advanced Linear Programming enforces absolute privacy compliance while mathematically minimizing distortion.



Three coordinated phases turn raw data into audit-ready intelligence.



A robust TUI eliminates pure-CLI friction.

Validation-first, with live diagnostics and instant overrides.



Validation-first

Catches nulls, schema errors, and peer-constraint breaches before any heavy run.



Intuitive config

Dropdown selection of entities, metrics, and dimensions.

The screenshot shows the 'Privacy-Compliant Peer Benchmark' application window. It features three main configuration panels on the left and a status/log panel on the right.

- 1 • Data Source:** Includes a 'CSV file path' field with a 'Browse...' button and a text input showing 'D:\Projects\autobench\tests\fixtures\gate_de'. Below, it displays file details: 'gate_demo.csv • 8 columns • 1.7 KB' and a list of columns: 'issuer_name, year_month, card_type, channel, txn_cnt, total...'.
- 2 • Entity:** Contains two dropdown menus: 'Entity ID column' (set to 'issuer_name') and 'Target entity (blank = peer-only)' (set to 'Target').
- 3 • Analysis Options:** Includes a 'Time period column (optional)' dropdown (set to 'year_month'), an 'Output filename' text input (set to 'C:\Users\ei176097\AppData\Lo...'), and an 'Optimization preset' dropdown (set to 'balanced_default') with a 'Preset Guide' button. Below these are several checkboxes, all of which are checked: 'Analyze impact', 'Compare presets', 'Validate input', 'Include calc. metrics (CSV)', and 'Acknowledge accuracy-firs'. A descriptive text at the bottom reads: 'Standard Analysis - Balance of compliance and consistency (formerly standard)'.
- Run Status:** Shows a 'Ready' status with 'Modeshare' and 'Presetbalanced_default' details.
- Last Run:** Displays 'No runs yet.'
- Execution Log:** A scrollable log showing timestamps and status messages: '2026-06-16 09:54:42,638 - INFO - Loaded 6 preset', '2026-06-16 09:54:42,639 - INFO - Loaded preset:', '2026-06-16 09:54:42,639 - INFO - Initialized Con', and '2026-06-16 09:54:42,639 - INFO - Applied preset:'.

The bottom of the window features a command bar with shortcuts: 'Open CSV', 'Run f1', 'Preset Guide ^a', 'Advanced ^e', 'Export Adv ^L', 'Clear Log', and a 'palette' button.



Real-time diagnostics

Integrated log handling keeps the UI responsive during background work.



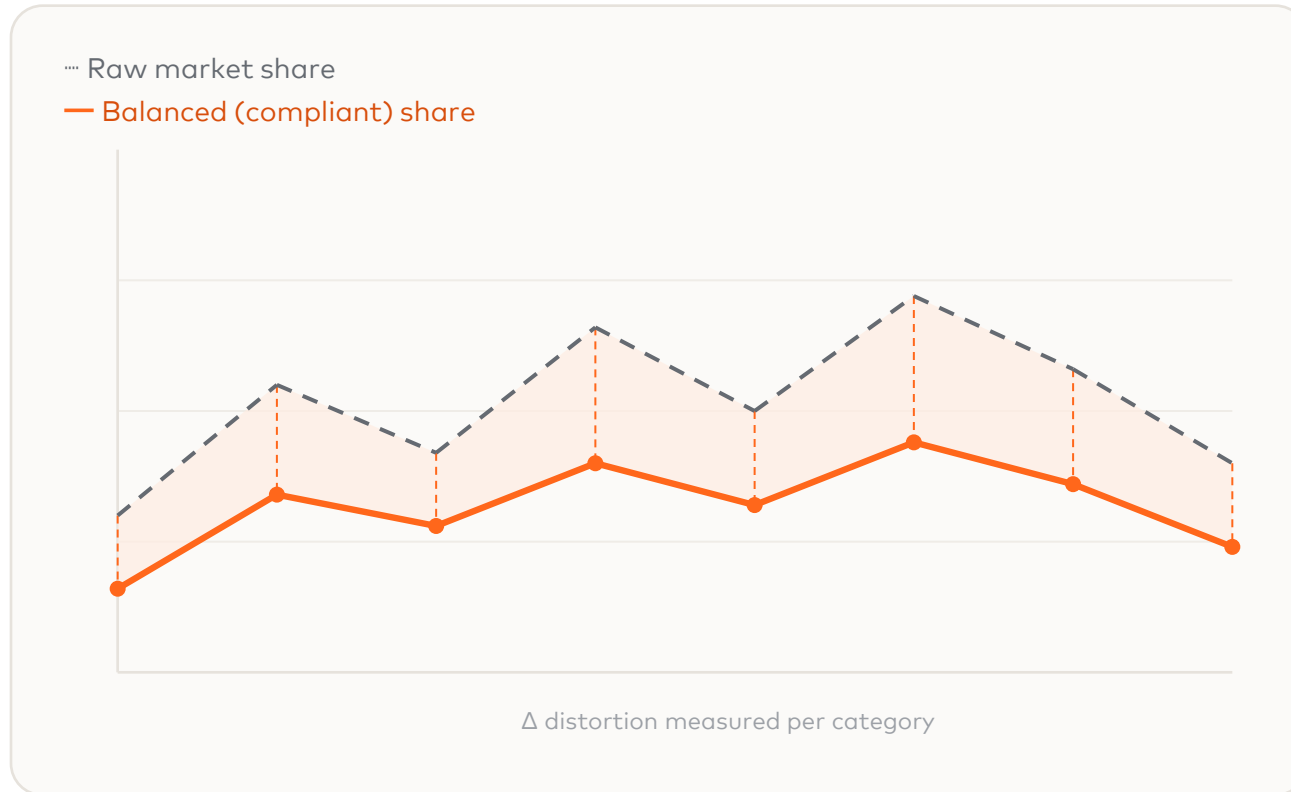
Instant overrides

Toggle advanced panels (Ctrl+A) to inject custom rules on the fly.



We quantify the true cost of meeting privacy caps.

Every percentage point of distortion is measured, not assumed.



Distortion Analysis

Computes the exact percentage-point delta between raw market truth and compliant reporting.



Rank-Change Tracking

A dedicated sheet records every rank shift before and after weighting.



Preset Comparison

--compare-presets builds a matrix of mean distortion to find the optimal configuration.

Insight. We don't just enforce rules, we quantify their mathematical impact.



The system actively guards against unauthorized data combinations.

DATA SCENARIO	POLICY CONDITION	STATUS
 Fraud / Chargebacks	Must set privacy_basis: clearing_spend	ENFORCED
 Digital Wallets	Requires explicit Privacy-team review	MANUAL APPROVAL
 Dual-Entity-Axis	Requires explicit Privacy-team review	MANUAL APPROVAL
 Top-Merchant Lists	Strictly prohibited	HARD BLOCKED



Actionable intelligence, packaged for decisions and for audit.



.xlsx

The Excel Report

- Executive summary & run metadata.
- Category comparisons (Target vs. Best-in-Class).
- Rank changes & weight methods.



.CSV

The Balanced Export

- BI-ready for Tableau & Power BI.
- Enriched metrics (Distortion_PP, Raw_Share).
- Schema-validated for parity with Excel.



.zip

The Audit Package

- Immutable snapshot of inputs, config & logs.
- Ready for immediate regulatory submission.

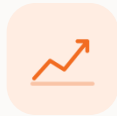


Autobench turns regulatory constraint into analytical advantage.



Absolute Compliance

- 100% adherence to Control 3.2 privacy caps.
- Cryptographically verifiable audit trails.



Mathematical Superiority

- Industry-leading LP solvers minimize data distortion.
- Unlocks insight in sparse data that legacy tools abandon.



Operational Agility

- TUI and lean mode democratize access at massive scale.
- No infrastructure bloat.

Compliant data is no longer a compromise. It is an exact science.



Automating the balancing phase cuts benchmark time by ~75%.

Data extraction stays flat, every gain comes from eliminating manual, senior-dependent analysis.

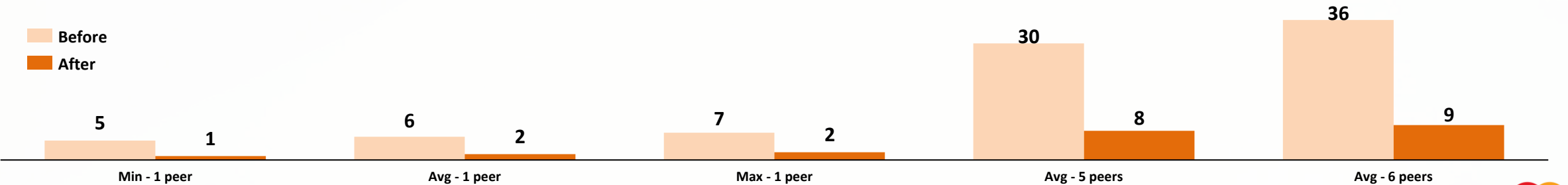
Balancing Time Per Peer — Before vs. After			
Metric	Before	After	Improvement
Minimum	5 days	1 day	4 days
Average	6 days	1.5 days	4.5 days
Maximum	7 days	2 days	5 days

Average reduction: ~75% in balancing time per peer across all project sizes.

Market-Wide Scalability — Project Time Comparison			
Project Size	Before	After	Time Saved
5 peers	30 days	7.5 days	22.5 days
6 peers	36 days	9 days	27 days

Large benchmarking studies that previously required **multiple analysts and weeks of manual work** can now be executed in **roughly one week, by a single analyst**.

Visual demonstration of time savings using the tool



A&F Dashboard — scalable benchmarking for the LAC market.



PEERS

Ready-to-Use Peer Groups

- Pre-defined peer groups for issuers and acquirers.
- 14 months of history (Jan '25 – Feb '26).
- Brazil live; LAC rollout underway.



METRICS

Deep Performance Breakdowns

- Raw & clean auth rate, fraud rate, declined-reason share.
- Count & amount, by credit/debit, CP/CNP, dom/cross-border.
- Tokenization, 3DS, MCC, ticket size & more.

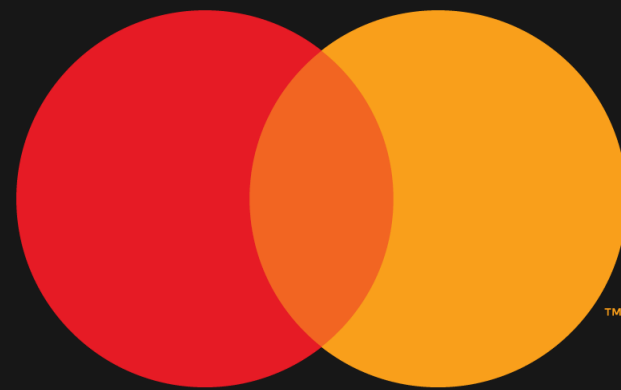


IMPACT

Value to ACS

- Cuts effort on the hardest steps: extraction & peer grouping.
- Frees senior time for higher-margin analysis.
- Unlocks "A&F Report as a Service" revenue.





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